Step 4 — এখন তোমার wgs_df দেখো

```
display(wgs_df)
```

	file_id	file_name	project	patient	data_type	data_format	experimental_strategy	access	file_size_GB
0	51597c61-0d49-4b0b-a05c-a2573a338de0	725b35d1-7aaa-4471-9539-6cc0ea98a6de_wgs_gdc_r...	None	None	Aligned Reads	BAM	WGS	controlled	23.775
1	b852dfaa-8923-4ebf-bf42-22e84ddb23a0	TCGA-COAD.7ff10995-0394-45bc-8caa-6a7ed9cad562...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.000
2	7ad18a0c-d34b-4d0e-9fcc-cd766e2d2a84	d13f4522-90dc-41c8-a274-fbd9c0cd2e60_wgs_gdc_r...	None	None	Copy Number Segment	TXT	WGS	open	0.000
3	7c9557b0-b0ed-4c17-a758-b0d6af6192b0	a9ea9348-2251-4b93-aac3-8faa8e9a9b68.wgs.gatk4...	None	None	Intermediate Analysis Archive	TAR	WGS	controlled	0.262
4	a7338c67-eb5d-4034-8ccf-b3519322c784	TCGA_COAD.e02120fa-895c-48e4-ad7e-d6df10eda491...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.014
...	...	...	...	...	...	...	...	...	...
95	86cb24e7-90a3-4234-a61f-ef5375a0b44d	ea65ef23-6d0f-4f89-9026-cb254379fec9_wgs_gdc_r...	None	None	Aligned Reads	BAM	WGS	controlled	288.802
96	f6beb916-7d03-4411-b744-8a0babfe887f	TCGA_COAD.43a4ae71-08c8-42f5-954f-af4f95da028d...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.112
97	16ff61dcd-e3f4-4e92-9754-ba4b6a0bb8dd	TCGA_COAD.0df064bb-ed15-42f9-99ba-37a921ea43dd...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.084
98	c958ec20-ef57-4a5c-a633-4e67e5bbc02d	c65b8095-f2f8-42ad-bda6-6c4f2738bf54_wgs_gdc_r...	None	None	Aligned Reads	BAM	WGS	controlled	39.668
99	eb7e8a12-6f75-4eb6-9326-e3f20ef6775f	TCGA_COAD.77824e79-5296-4f34-959f-b8a1b0c5b5af...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.000

100 rows × 9 columns

Step 5 — Data type দেখো

```
print("Data types found:")
display(
    wgs_df["data_type"]
    .value_counts()
    .to_frame("number_of_files")
)
```

Data types found:

data_type	number_of_files
Structural Rearrangement	26
Raw Simple Somatic Mutation	23
Annotated Somatic Mutation	22
Aligned Reads	15
Intermediate Analysis Archive	7
Copy Number Segment	7

Step 6 — COAD এবং READ আলাদা করে দেখো

Step 6 — COAD এবং READ আলাদা করে দেখো

```
print("TCGA-COAD")
display(
    wgs_df[wgs_df["project"] == "TCGA-COAD"]
)

print("\nTCGA-READ")
display(
    wgs_df[wgs_df["project"] == "TCGA-READ"]
)
```

TCGA-COAD

file_id	file_name	project	patient	data_type	data_format	experimental_strategy	access	file_size_GB

TCGA-READ

file_id	file_name	project	patient	data_type	data_format	experimental_strategy	access	file_size_GB

Step 7 — সবচেয়ে গুরুত্বপূর্ণ: mutation-related file খুঁজো

```
mutation_keywords = [
    "mutation",
    "variant",
    "variation"
]

mutation_df = wgs_df[
    wgs_df["data_type"]
    .fillna("")
    .str.lower()
    .apply(
        lambda x: any(
            keyword in x
            for keyword in mutation_keywords
        )
    )
].copy()

print("Mutation-related WGS files:", len(mutation_df))

display(mutation_df)
```

Mutation-related WGS files: 45

	file_id	file_name	project	patient	data_type	data_format	experimental_strategy	access	file_size_GB
1	b852dfaa-8923-4ebf-bf42-22e84ddb23a0	TCGA-COAD.7ff10995-0394-45bc-8caa-6a7ed9cad562...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.000
4	a7338c67-eb5d-4034-8ccf-b3519322c784	TCGA_COAD.e02120fa-895c-48e4-ad7e-d6df10eda491...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.014
5	27475108-43d8-4519-872e-4bb716bf8825	62d68c3e-602d-4efa-aaa9-80a109818175.svaba.som...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.001
6	546261a0-7318-4e7b-b551-0bbf8e58ff4d	TCGA-COAD.7fa71b99-1560-47ff-8a04-b78ef7d0dc0a...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.002
10	d32ab4ba-965c-46c5-aaeb-3a4a0bed2fa5	TCGA_COAD.b07c2bcd-5a1c-4234-a4db-09a94635ff5c...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.000
11	e3b6143e-05ad-4573-8d7a-a7a0dfcb1def	TCGA_COAD.f53fab29-fcf1-413e-bc0e-c82e7b73d4cf...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.007

bles Terminal

```
display(mutation_df)
```

```
Mutation-related WGS files: 45
```

	file_id	file_name	project	patient	data_type	data_format	experimental_strategy	access	file_size_GB
1	b852dfaa-8923-4ebf-bf42-22e84ddb23a0	TCGA-COAD.7ff10995-0394-45bc-8caa-6a7ed9cad562...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.000
4	a7338c67-eb5d-4034-8ccf-b3519322c784	TCGA_COAD.e02120fa-895c-48e4-ad7e-d6df10eda491...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.014
5	27475108-43d8-4519-872e-4bb716bf8825	62d68c3e-802d-4efa-aaa9-80a109818175.svaba.som...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.001
6	548261a0-7318-4e7b-b551-0bbf8e58ff4d	TCGA-COAD.7fa71b99-1560-47ff-8a04-b78ef7d0dc0a...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.002
10	d32ab4ba-965c-46c5-aaeb-3a4a0bed2fa5	TCGA_COAD.b07c2bcd-5a1c-4234-a4db-09a94635f55c...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.000
11	e3b6143e-05ad-4573-8d7a-a7a0dfcb1def	TCGA_COAD.f53fab29-fcf1-413e-bc0e-c82e7b73d4cf...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.007
14	6013ce86-980b-4aca-88fb-8828c25623d7	3f3a97bf-c40d-42c3-b318-cb0b5c948bc5.svaba.som...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.000
16	87d7633a-3ac6-4904-aeb7-016c02344862	TCGA_COAD.dbe5bd11-ba35-44d7-a7ff-7645652bf73a...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.005
17	2ab243e3-e54a-4b44-a174-6fee1be100d2	TCGA_COAD.9817eaa6-9e7d-4c96-8abb-8e31acb735bb...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.004
22	5f9cc0f5-7576-414a-825f-4c1cc040be2d	TCGA-COAD.d442d243-dbdd-4893-a51e-5bb80dd1f8c7...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.003
23	53954feb-8ca5-4cf7-8dd8-f31d965aa149	364c1106-664f-4378-9da3-9f527d7a67c5.svaba.som...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.000
25	7b250026-e574-4aa5-9265-bc50d461509b	TCGA-COAD.fb0b90a8-8d50-4472-85c7-866caebaa7cc...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.002
27	5abe5cb8-d17a-4188-8f03-5e223ec9ee2b	TCGA_COAD.ba72c9f6-7a73-4333-9793-b1e9be10f9aa...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.004
28	31e821a5-2fcc-4f34-b8f6-ca4e59bb1829	TCGA-COAD.78cac9dd-af9e-4bdf-a796-e4c823dfa845...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.002
32	72832a32-2a27-47df-854b-705eacca3e2d	TCGA_COAD.c170a7e0-270a-4cda-8b68-69e65d829608...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.001
33	8f5c6dc9-432c-43f0-9891-10ac873d4532	TCGA_COAD.174df798-1e0e-4f92-a782-248e78136cf1...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.005
34	e3a48221-3e37-481e-b3a4-237085af636e	647b0944-00cc-40a8-a58d-f616d77f5439.svaba.som...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.000
35	35d7ad55-af3b-40b5-89cb-66bfc074a566	TCGA-COAD.f55704be-b313-4c9d-9add-7100776a01b9...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.002
41	5bad2aeb-3141-4367-9b58-6aee7e6972ed	TCGA-COAD.6aab2bbb-6401-4d9e-8b8e-1cf4167a46b6...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.001
43	d1cbad27-9179-468d-911e-2ff179e08ffa	TCGA_COAD.0f07392b-b102-43d1-8c6d-d0e1551252d5...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.004
46	5547334b-8bf4-4e8e-be9d-43e7eb2a05f3	TCGA_COAD.8a3bc978-d3a9-4795-a1b3-21a46725fd45...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.102
47	1e42d572-64ab-4b3f-98c5-12a6bcc158f3	TCGA-COAD.36857a13-3597-4505-87bb-a073dd85a181...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.034
51	bf4544a7-1035-4a93-8f94-8fab04b77d6a	TCGA-COAD.43d35cfe-94e8-4744-adf1-40bcf2836791...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.030
53	0af3b6f2-608f-4e3a-839b-4d0cd3f78778	TCGA_COAD.36133b51-3347-41d1-917f-b5092736dd8b...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.035
54	56cab21d-082a-4030-93af-3bef44609efa	TCGA_COAD.6c5979b1-c06c-4484-8885-6a338ec63be7...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.076
55	7f12d5ce-e9ae-485f-8bc9-be6f1ed84631	9655d0b7-4569-4be7-b1b9-4eb630efb6c4.svaba.som...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.013
63	289e64f2-dcf2-4593-b6f5-21980da89d56	e8e65f93-d79f-42c9-963f-c1a970116e43.svaba.som...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.000
65	fb84801e-d62f-4d58-b602-eb79abe4cf93	TCGA-COAD.27a5c5a1-1034-4862-be77-0a3a2ebc3b5b...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.002
66	63e7c888-bc06-4e4e-a5fb-ad71a3f43240	TCGA-COAD.df1f636f-7882-4d72-8d3e-f2a8a840975c...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.000
67	80572ffb-e140-4000-82db-f0f68e9c3071	TCGA-COAD.8e04da69-2232-41c2-8a7c-42ec8de09753...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.002

```
display(mutation_df)
```


28	31e821a5-2fcc-4f34-b8f6-ca4e59bb1829	TCGA-COAD.78cac9dd-af9e-4bdf-a796-e4c823dfa845...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.002
32	72832a32-2a27-47df-854b-705eacca3e2d	TCGA_COAD.c170a7e0-270a-4cda-8b68-69e65d829608...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.001
33	8f5c6dc9-432c-43f0-9891-10ac873d4532	TCGA_COAD.174df798-1e0e-4f92-a782-248e78136cf1...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.005
34	e3a48221-3e37-481e-b3a4-237085af636e	647b0944-00cc-40a8-a58d-f616d77f5439.svaba.som...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.000
35	35d7ad55-af3b-40b5-89cb-66bfc074a566	TCGA-COAD.f55704be-b313-4c9d-9add-7100776a01b9...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.002
41	5bad2aeb-3141-4367-9b58-6aee7e6972ed	TCGA-COAD.6aab2bbb-6401-4d9e-8b8e-1cf4167a46b6...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.001
43	d1cbad27-9179-468d-911e-2ff179e08ffa	TCGA_COAD.0f07392b-b102-43d1-8c6d-d0e1551252d5...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.004
46	5547334b-8bf4-4e8e-be9d-43e7eb2a05f3	TCGA_COAD.8a3bc978-d3a9-4795-a1b3-21a46725fd45...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.102
47	1e42d572-64ab-4b3f-98c5-12a6bcc158f3	TCGA-COAD.36857a13-3597-4505-87bb-a073dd85a181...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.034
51	bf4544a7-1035-4a93-8f94-8fab04b77d6a	TCGA-COAD.43d35cfe-94e8-4744-adf1-40bcf2836791...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.030
53	0af3b6f2-608f-4e3a-839b-4d0cd3f78778	TCGA_COAD.36133b51-3347-41d1-917f-b5092736dd8b...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.035
54	56cab21d-082a-4030-93af-3bef44609efa	TCGA_COAD.6c5979b1-c06c-4484-8885-6a338ec63be7...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.076
55	7f12d5ce-e9ae-485f-8bc9-be6f1ed84631	9655d0b7-4569-4be7-b1b9-4eb630efb6c4.svaba.som...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.013
63	289e64f2-dcf2-4593-b6f5-21980da89d56	e8e65f93-d79f-42c9-963f-c1a970116e43.svaba.som...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.000
65	fb84801e-d62f-4d58-b602-eb79abe4cf93	TCGA-COAD.27a5c5a1-1034-4862-be77-0a3a2ebc3b5b...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.002
66	63e7c888-bc06-4e4e-a5fb-ad71a3f43240	TCGA-COAD.df1f636f-7882-4d72-8d3e-f2a8a840975c...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.000
67	80572ffb-e140-4000-82db-f0f68e9c3071	TCGA-COAD.8e04da69-2232-41c2-8a7c-42ec8de09753...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.002

70	25a4ba8b-914c-40ac-b8fc-d227529e7562	TCGA_COAD.214a6929-94b8-49a3-a1c4-d017f6e8d964...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.000
71	3495d74c-4875-4d59-9938-a13308ba6774	TCGA_COAD.bc8e0da7-cd94-443d-a31a-43f6fd5e2954...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.001
75	faddfe27-d561-48b3-a433-3dbc4213c9a3	TCGA_COAD.bb520b61-e401-4c18-9e45-255c32d9532c...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.000
76	3e61af98-6acc-451e-a55c-e853b6bed2c2	0b97f175-1966-48a2-8f60-5ae56f284065.svaba.som...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.000
80	001bc942-af37-418e-affd-61d09da6b7b1	TCGA-COAD.8f3833e0-53b1-4f4e-b57b-44e15b3c502f...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.000
81	6edd6492-b54c-4bbe-be99-0ca2fcfbec65	TCGA_COAD.9b089b87-ce46-4ee3-a0a8-44326ccc967b...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.007
82	3f26d23e-e3f8-40b7-a5a6-8179378571a4	TCGA_COAD.cd3566d1-83f8-41b2-9aea-e66e66b3c22b...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.000
83	b4e22dfa-ae89-43bd-94b0-ccc6bfc71170	TCGA_COAD.1fe962b2-c1f5-4159-8f16-427173050afa...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.006
85	f0901257-70e5-45ff-88a2-ea95f4581988	TCGA-COAD.0b6e7125-b4ad-44e3-bcd4-f62a2832e2f7...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.032
86	00aa66cf-be90-41f3-9aad-986031b83fd2	TCGA_COAD.30230f50-5d9c-4fdc-934e-7de64826f94e...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.043
88	f74b8569-549f-4d64-aef7-a9216d47bcf0	ed79fc95-e836-4f60-8e8b-b37b59815e10.svaba.som...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.015
89	67c603c4-192f-4c4e-9833-6e2d1b9a430f	TCGA-COAD.cc6f438e-428e-472c-83d0-49b0b7549441...	None	None	Raw Simple Somatic Mutation	VCF	WGS	controlled	0.038
96	f6beb916-7d03-4411-b744-8a0babfe887f	TCGA_COAD.43a4ae71-08c8-42f5-954f-af4f95da028d...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.112
97	16f61dcd-e3f4-4e92-9754-ba4b6a0bb8dd	TCGA_COAD.0df064bb-ed15-42f9-99ba-37a921ea43dd...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.084
99	eb7e8a12-6f75-4eb6-9326-e3f20ef6775f	TCGA_COAD.77824e79-5296-4f34-959f-b8a1b0c5b5af...	None	None	Annotated Somatic Mutation	VCF	WGS	controlled	0.000